

Emergency Vehicle Sensing and Communication Network (ISAC-V2X)

An ISAC-V2X Framework for Traffic-Density Sensing and Emergency Vehicle Alerting

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Abstract—Emergency vehicles can lose valuable response time when their routes pass through congested road segments, while civilian drivers may become aware of an approaching emergency vehicle only at short range. This project proposes an Integrated Sensing and Communication (ISAC) based Vehicle-to-Everything (V2X) framework in which a shared radio signal is used both to estimate traffic density and to carry emergency-vehicle communication. The proposed system uses roadside and relay nodes to count active beacon signals as a coarse traffic-density proxy. Live density information is then supplied to a route-optimization module based on Dijkstra or A* implemented in software to select a less-congested path. The emergency vehicle subsequently broadcasts a targeted “make way” alert over a software-defined-radio (SDR) link to vehicles on the selected route. The project is planned around a custom OFDM physical layer implemented using GNU Radio and a USRP B210 SDR, with virtual relay nodes supported through ZeroMQ/UDP. Our Milestone 1 submission establishes the problem definition, reviews three representative benchmarks—HAAS Alert, DSRC-based rear-end collision warning, and ISAC-enabled vehicular networks—and identifies the project gap at the intersection of RF-based congestion sensing and emergency V2X alerting. We also define measurable evaluation targets, system scope, assumptions, and the roadmap for prototype development and end-to-end validation of the project.

Index Terms—Integrated sensing and communications, ISAC, V2X, emergency vehicle, DSRC, software-defined radio, GNU Radio, USRP B210, OFDM, traffic density, route optimization.

I. INTRODUCTION

Emergency response systems depend strongly on the ability of an emergency vehicle (EV) to move through traffic without unnecessary delay. In congested corridors, the EV driver may not have sufficient information about the density of individual road segments to select the fastest route. At the same time, civilian vehicles may not receive an early warning that an EV is approaching their road segment. These two issues create complementary problems: the emergency vehicle has limited knowledge of the traffic state ahead, while nearby civilian vehicles have limited awareness of the approaching emergency vehicle.

Existing emergency-vehicle warning systems demonstrate the value of digital alerts. The HAAS Alert system, for example, provides responder-to-vehicle (R2V) and responder-to-responder (R2R) alerts using real-time vehicle information and cloud-based processing [1]. Its evaluation by the U.S. Department of Homeland Security’s National Urban Security Technology Laboratory reported that the technology could be a valuable tool for reducing collisions involving emergency response vehicles [1]. However, the architecture depends on connected vehicles, navigation applications, GPS, cellular communication, and cloud processing.

Dedicated Short-Range Communication (DSRC) has also been demonstrated for vehicle-to-vehicle safety applications. Zhao *et al.* developed a DSRC-based rear-end collision warning system and evaluated it using controlled field tests [2]. This line of work demonstrates that direct radio communication can support timely vehicle-safety warnings, but the communication channel itself does not provide the traffic-density sensing and dynamic route-selection functionality targeted in this project.

The other major family of standards for V2X communication is V2X-LTE or C-V2X. Abbasi *et al.* established that the performance of C-V2X is generally superior to

DSRC/802.11p for distances greater than 100 meters [3]. For our project we will remain cognizant of this limitation in terms of the design and the distance our nodes may be placed. The relay node mechanism may also help to mitigate this distance/performance tradeoff.

More recent work on ISAC-enabled vehicular networks considers using the same radio resources for both sensing and communication. Du *et al.* describe an ISAC framework for V2I networks in which sensing and communication are jointly designed [4]. The present project adopts the same high-level principle but targets a smaller and more accessible implementation: coarse traffic-density sensing and emergency alerting using SDR hardware rather than mmWave massive-MIMO infrastructure.

II. PROBLEM DEFINITION

The project problem is defined by three related operational challenges:

A. Emergency-vehicle response latency

An EV may lose critical response time when it enters a congested road corridor. Without a live estimate of the density of alternative road segments, the driver or routing system cannot reliably determine whether the current route remains preferable.

B. Route congestion blindness

The EV requires a mechanism that converts radio observations into a useful estimate of traffic density. The proposed project does not attempt to reconstruct the complete physical traffic state. Instead, it uses the number of active beacon signals observed on a road segment as a low-cost density proxy. This proxy is then used to update the cost of the corresponding road segment in a routing graph.

C. Belated civilian awareness

Civilian drivers travelling along the EV’s selected route may become aware of the emergency vehicle only when visual or audible warning cues become noticeable. The proposed system therefore includes a direct RF broadcast mechanism through which the EV can send a targeted “make way” alert to nearby nodes.

D. Core problem statement

The project seeks to design and validate a lightweight two-way RF system in which a shared physical link supports both traffic-density sensing and V2X communication. The intended closed loop is:

$$\begin{aligned}
 \text{EV intent} &\rightarrow \text{RF density sensing} \\
 &\rightarrow \text{route selection} \\
 &\rightarrow \text{RF alert broadcast.}
 \end{aligned} \tag{1}$$

The shared-channel system is the centerpiece of our project.

III. PROPOSED SYSTEM

A. System concept

The proposed architecture consists of three functional modules: a density-sensing module, a route-optimization module, and a broadcast-alert module. The roadside-unit (RSU) and relay nodes will count active beacon signals per road segment as a rough heuristic for density. The resulting live density information will be used by Dijkstra or A* to dynamically calculate a route, after which the EV will transmit a "make way" alert using GNU Radio and USRP hardware.

B. Density sensing module

The density module is based on beacon counting. Each road segment is associated with one or more sensing/relay nodes. The number of active beacon signals observed during a measurement interval is treated as a proxy for the number of vehicles present on that segment. This approach is deliberately lightweight: the project does not require a dedicated radar sensor or detailed object classification at the Milestone 1 stage.

The density proxy will later be validated against manually counted ground-truth traffic in at least two laboratory test scenarios; this is one of our main objectives.

C. Route optimization module

The routing module converts the measured density of each road segment into a dynamic edge cost. The project plan identifies Dijkstra and A* as the candidate routing algorithms. The selected algorithm will recompute the emergency vehicle path when live density information changes.

The routing layer is intentionally separated from the RF physical layer so that it can first be tested using synthetic or recorded density values. This modular structure reduces the difficulty of debugging the complete system and permits independent verification of the routing logic.

D. Broadcast alert module

After a route is selected, the EV transmits a targeted alert to nearby vehicles or relay nodes on the selected path. The project specifies GNU Radio and USRP hardware for this module and intends to demonstrate an RF broadcast channel reaching at least three receiving nodes with a packet delivery ratio (PDR) of at least 90%.

E. Physical-layer concept

The proposed physical layer is a custom OFDM architecture inspired by IEEE 802.11p-style vehicular communication (ideally we would provide an implementation conforming to the standard, but regulatory details of transmission in the 5.9Ghz band are unclear). The selected hardware is the USRP B210 SDR. We shall use a 5.8 or 5.9 GHz operating frequency, GNU Radio and Python software, and ZeroMQ/UDP sockets for interconnecting virtual relay nodes.

IV. STATE OF THE ART AND RELATED WORK

Milestone 1 uses three representative benchmarks to position the proposed system: HAAS Alert, DSRC-based rear-end collision warning, and ISAC-enabled vehicular networks. The comparison is summarized in Table I.

A. HAAS Alert

The HAAS Alert report describes two main use cases: responder-to-vehicle (R2V) alerts for civilian drivers and responder-to-responder (R2R) alerts between equipped emergency vehicles [1]. In the R2V case, alerts are associated with the predicted path of an emergency vehicle and can be delivered through navigation applications or vehicle systems. In the R2R case, the system identifies potential collisions between emergency vehicles connected to the HAAS Safety Cloud.

The system therefore establishes an important benchmark for the alerting component of the proposed project. Its evaluation also demonstrates that operational suitability includes more than communication accuracy: alert timing, usability, configuration, reporting, and operator interaction are relevant considerations.

B. DSRC-based rear-end collision warning

Zhao *et al.* proposed a DSRC-based rear-end collision warning system that explicitly considers transmission delay and GPS error in the safety-distance model [2]. The system was assembled using off-the-shelf DSRC and mobile technology and evaluated through controlled test-track scenarios.

This work serves as evidence that direct DSRC/802.11p-style communication can support vehicle safety warnings. The important distinction for the present project is that the proposed system uses the RF channel not only for communication but also for sensing traffic density and supporting route selection.

C. ISAC-enabled vehicular networks

Du *et al.* describe ISAC as a joint design in which sensing and communication share radio resources and can provide both integration and coordination gains [4]. Their framework discusses V2I networks, sensing-assisted communication, beam management, and the use of a common waveform for sensing and communication.

The present project follows this principle at a smaller implementation scale. Instead of attempting high-resolution localization or mmWave beam tracking, the proposed system uses a coarse density proxy that can be implemented using accessible SDR hardware. This makes the project suitable for laboratory-scale validation while preserving the central ISAC idea.

V. RESEARCH GAP AND PROJECT NOVELTY

The literature reviewed in Milestone 1 reveals three partially separated solution spaces. First, emergency warning platforms such as HAAS Alert demonstrate effective digital warning and responder-to-responder communication. Second, DSRC

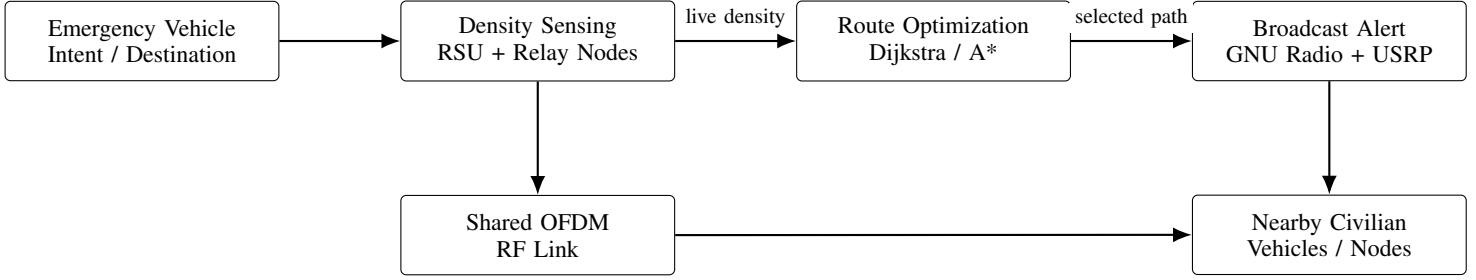


Fig. 1. Proposed ISAC-V2X system architecture showing the density sensing, route optimization, and broadcast alert modules.

TABLE I
SUMMARY OF RELATED WORK AND PROJECT POSITIONING.

Benchmark	Main capability	Strength	Limitation relative to this project
HAAS Alert [1]	Responder-to-vehicle and responder-to-responder emergency alerts using real-time vehicle data, GPS, cellular connectivity, cloud processing, and user-facing applications.	Demonstrates an operational emergency-vehicle warning concept and evaluates functionality, reliability, usability, and suitability.	Does not combine RF traffic-density sensing with the alerting loop; the architecture is cloud and connectivity dependent.
Zhao <i>et al.</i> [2]	DSRC-based rear-end collision warning using direct V2V communication and an error-component safety-distance model.	Provides a field-tested example of direct RF safety communication and considers communication delay and GPS error.	Primarily a communication and collision-warning system; it does not provide the proposed RF density sensing, dynamic routing, or EV-specific closed loop.
Du <i>et al.</i> [4]	ISAC-enabled V2I framework using shared sensing and communication resources, including sensing-assisted beam management.	Establishes the principle of integrating sensing and communication rather than treating them as separate functions.	Targets advanced mmWave/massive-MIMO vehicular systems, whereas this project focuses on coarse density sensing using accessible SDR hardware.
Proposed ISAC-V2X	RF-based density proxy + dynamic routing + emergency vehicle broadcast alert over a shared OFDM link.	Combines sensing, route selection, and V2X alerting into one lightweight closed-loop prototype.	Milestone 1 defines the architecture; quantitative system performance will be established during later prototype and evaluation milestones.

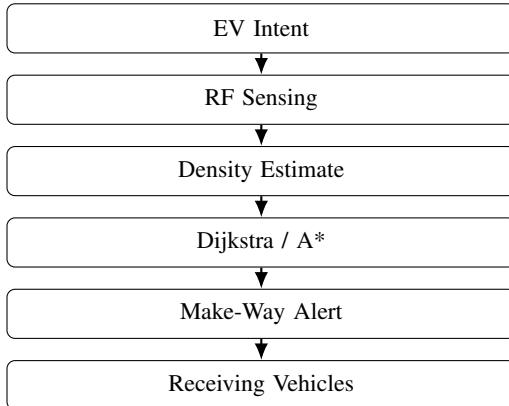


Fig. 2. Closed-loop operating sequence of the proposed ISAC-V2X system.

safety research demonstrates direct V2V communication for collision-warning applications. Third, ISAC research demonstrates that sensing and communication can be jointly designed using shared radio resources.

The project gap identified from these benchmarks is the lack of a lightweight framework that combines:

- 1) RF-based traffic-density sensing,
- 2) dynamic emergency-vehicle route selection,
- 3) direct V2X emergency alert broadcasting, and
- 4) a shared RF/OFDM communication platform.

The proposed novelty is therefore not a claim of inventing ISAC or V2X communication independently. Rather, the contribution is the integration of these functions into a laboratory-scale emergency-vehicle system using accessible SDR hardware. We identify the combination of RF-based congestion sensing and V2X emergency alerting as the targeted research gap.

VI. SCOPE, ASSUMPTIONS, AND CONSTRAINTS

A. Scope

The scope finalized for the project is:

- develop a custom OFDM-based RF communication link in GNU Radio;
- use USRP B210 hardware for the primary RF node;
- use beacon counting as a coarse traffic-density proxy;
- validate the density proxy against manual ground truth;
- implement Dijkstra/A* based dynamic route selection;

TABLE II
PROJECT EVALUATION CRITERIA.

Criterion	Target / assessment method
RF reliability	Demonstrate at least 90% PDR across 3 or more receiving nodes.
Density proxy validity	Compare beacon-count estimates with manual ground truth in at least two laboratory scenarios.
Route adaptation	Verify that the Dijkstra/A* implementation recomputes the selected path within a reasonable window when live density values change.
End-to-end operation	Demonstrate EV intent → sensing → routing → alert broadcast.
Signal identification	Measure false-positive and false-negative behaviour of the EV broadcast identifier.
Latency	Compare alert latency and time saved for sensing-enabled and baseline cases.
Implementation feasibility	Verify that the architecture can be implemented using GNU Radio, Python, USRP B210, and virtual nodes.

- transmit emergency “make way” alerts to vehicles/nodes on the selected route;
- support at least three receiving nodes in the RF demonstration;
- identify EV alert signals using a preamble/header and evaluate false-positive and false-negative behaviour;
- integrate the modules into an end-to-end laboratory demonstration.

B. Assumptions

For Milestone 1, the following assumptions are made:

- road segments can be represented as edges in a graph;
- the number of active beacon signals provides a useful coarse proxy for density;
- density measurements can be updated frequently enough for route recomputation;
- RF communication can be tested using a controlled laboratory topology;
- virtual relay nodes can be used when physical nodes are insufficient for early testing;
- the project’s functionality is evaluated as a prototype rather than a production-certified emergency response system.

C. Constraints

The major constraints are the limited number of SDR hardware nodes, laboratory-scale RF conditions, coarse density estimation, and the need to separate software-module validation from end-to-end RF validation. The proposed system should therefore be interpreted as a proof-of-concept framework. Real-world deployment would require additional regulatory, reliability, cybersecurity, interoperability, and safety validation that are outside the Milestone 1 scope.

VII. CRITERIA FOR ASSESSING THE PROPOSED SYSTEM

The project requires measurable criteria so that later milestones can determine whether the proposed architecture is viable. Table II defines the principal criteria.

Our core objectives include PDR, density-proxy validation, dynamic routing, end-to-end integration, and signal identification. Performance comparisons of sensing-enabled and non-sensing cases are also included among the expected deliverables.

VIII. RESEARCH METHODOLOGY

Milestone 1 is primarily a design and research milestone. Our methodology therefore focused on establishing a traceable link between the identified problem, the literature, the proposed architecture, and the measurable criteria.

A. Literature review

The literature review uses four benchmark categories. The HAAS Alert report was used to understand an operational emergency-vehicle warning architecture and its evaluation dimensions. The DSRC study was used to understand direct RF vehicle-safety communication and the role of transmission delay and positioning errors. The ISAC review was used to establish the architectural principle of jointly using radio resources for sensing and communication. The C-V2X and DSRC performance analysis study was used to make decisions about the standard used for transmission and the performance limitations our system will have to work around.

B. System decomposition

The proposed system is decomposed into independent modules:

- 1) density sensing,
- 2) route optimization,
- 3) RF broadcast alerting,
- 4) signal identification, and
- 5) end-to-end integration.

This decomposition permits each software or DSP block to be tested independently before the final integration stage.

C. Density validation methodology

For each selected laboratory road-segment scenario, the number of manually counted vehicles will be recorded as the ground-truth value. The number of detected active beacon signals will be recorded over the same interval. The relationship between the two measurements will then be evaluated to determine whether the beacon count is sufficiently informative for the routing module.

We do not currently have a validated density estimator because the required experiments belong to the later implementation and evaluation stages.

D. RF communication methodology

The physical-layer development will use GNU Radio and a custom OFDM design. We shall use the USRP B210 as the principal SDR and ZeroMQ/UDP as the mechanism for connecting virtual relay nodes. Initial testing can therefore proceed with software-only nodes before increasing the number of physical RF nodes.

TABLE III
MILESTONE ROADMAP.

Phase	Focus	Expected output
Milestone 1	Problem and SOTA positioning	Problem definition, literature review, SOTA comparison, research gap, scope, assumptions, evaluation criteria, and finalized architecture.
Milestone 2	Module prototypes	Individual DSP/software prototypes for density sensing, Dijkstra/A* routing, and alert transmitter/receiver.
Milestone 3	System integration and demonstration	End-to-end integration using USRP B210 hardware and virtual relay nodes.
Final phase	Evaluation and report	PDR, latency, time-saved comparison, documented PHY parameters, signal-ID results, alert ruleset, logs, final documentation, and defense.

E. Route and alert validation

The route module will first be tested using generated density maps. Once the route logic is stable, it will receive live or recorded density updates. The selected route will then determine the intended alert recipients. The RF layer will be evaluated using packet delivery ratio, signal identification accuracy, and alert latency.

IX. MILESTONE PLAN AND EXPECTED DELIVERABLES

The project roadmap divides development into four phases. Milestone 1 is the current phase and focuses on problem definition, literature review, state-of-the-art positioning, and scope finalization.

The expected technical deliverables include GNU Radio TX/RX flowgraphs, a standalone density and routing module, a 3+ node SDR testbed, performance benchmarks, documented PHY specifications, signal-identification results, and threshold-based alerting logs.

X. CONCLUSIONS AND FUTURE OUTLOOK

Milestone 1 establishes the foundation for an ISAC-V2X emergency vehicle sensing and communication prototype. The problem has been divided into three operational challenges: emergency response latency, lack of live route-density information, and delayed civilian awareness. The proposed response is a closed-loop architecture that combines RF density sensing, dynamic route optimization, and direct emergency alert broadcasting.

The state-of-the-art review shows that existing approaches provide important individual capabilities. HAAS Alert demonstrates emergency-vehicle warning and responder-to-responder alerting, DSRC research demonstrates direct RF vehicle-safety communication, and ISAC research establishes the principle of shared sensing and communication. The proposed project combines these directions at a laboratory scale by using a coarse RF density proxy together with V2X emergency alerting.

The main outlook for subsequent milestones is that we ought to preserve the modular architecture. Density estimation, routing, RF transmission/reception, and signal identification

should first be validated independently before end-to-end integration. This will make it possible to determine whether limitations arise from the sensing proxy, routing logic, physical layer, or alerting mechanism.

At the end of Milestone 1, the project has a defined problem, a documented SOTA position, an identified research gap, a measurable evaluation plan, and a concrete future implementation path using GNU Radio, Python, USRP B210, custom OFDM signaling, and virtual relay nodes.

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